Group 1: Water Works

Testing Protocol (User and Unit Testing)

INFO 442

March 3rd, 2026

Key Features / Acceptance Tests:

Our acceptance tests / criteria are meant to see if our webapp meets the requirements of our users. We will be conducting user tests to roughly see if these acceptance criteria are met.

Key Feature 1: CSO Map and Notification Feature

- Map entries and context are understandable to the user
- Map explains CSO in language that people generally understand / care about
- Notification feature is simple to access

Key Feature 2: User/Onboarding Quiz

- Google authentication well integrated in sign up/log in page and user's email inputted into user database
- Quiz is completed in one sitting by the user
- User quiz responses are stored to the back-end

Key Feature 3: Your Impact Feature

- Users can easily navigate between cards
- The "consequences" and "alternatives" are useful information for the user

General

- Ensure that all pages and components load on screen for the user (navigation bar, cards, footer, headers, etc.)

Prototyping:

Because our web app's navigation isn't complex and we defined navigation within our static wireframe using annotations, we've decided to just use our static wireframes to begin coding.

We acknowledge that this could lead to some confusion between how features connect, but because the majority of our group shows up to class, we can discuss these issues among ourselves.

User Testing

User testing will be used to test if each key feature is useful for our targeted audience (provided user personas)

1. Motivate our user. Before testing, provide users with context about our project and problem statement
2. Allow users to take time to read over user personas, giving them space to imitate how that user persona would interact with the website
3. Provide them with a list of tasks to do as that user. This is the primary part of testing if these key features are useful.

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1. Over the spring and summer, you kayak very often. How would you consistently check if it is safe / sanitary to kayak throughout the season? (Intent: Test notification feature)
    a. Did you find the features we have useful for this task?
    b. What can we add or take away to improve the user experience?
2. You are unsure how to support the waters around you stay sanitary so that it can be enjoyed by you and the locals. How would you use Water Works to learn about how you can keep the waters clean? (Intent: Test if users find the “Your Impact Page” useful)
    a. Did you find the features we have useful for this task?
    b. What can we add or take away to improve the user experience?
3. Imagine that you have some habits that are bad for the environment / waterways that you wanted to break. How would you use Water Works to help with that? (Intent: Test how users navigate “Your Impact Page”)

    (General Intent: Test user roadmap, ensure that navigation’s intuitive)

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This information helped us realize that the language we were using didn’t coordinate with our intended audience (CSO, etc.). We expect that our fix and language will make the webapp intuitive for our users.

Unit Tests:

Our unit tests are meant to test the functionality of our web-app (verify that individual components work). Our intent is to use a **jest test** to conduct unit testing. **With the given time that we have, we will be adding tests to see if each component generally loads and some tests to test specific features.**

Key Feature 1: CSO Map and Notification Feature

- Ensure that notifications are sent and received by the user during a CSO event
- All points of cso location should be accurately shown in the map
- The points should have different color to show if there is a possible CSO event

Key Feature 2: User/Onboarding Quiz

- Google authentication well integrated in sign up/log in page and user's email inputted into user database
- Onboarding quiz data are inputted into user database

Key Feature 3: Your Impact Feature

- Cards are properly loaded on screen
- Both the "consequence" and "alternative" sections correspond with the card being clicked
- Users are able to navigate from one card to the other

General

- Ensure that all pages and components load on screen for the user (navigation bar, cards, footer, headers, etc.)

We expect that at the very minimum, all of our components load and users can interact with them. We also expect that our CSS styling allows for a bug-free experience in both desktop and mobile.

Identifying and Re-testing Bugs:

To decide which bugs to fix first, we'll label bugs by severity and based on our functional requirements

Severity: How significantly does the bug impact our web-app in its entirety? If we face a bug that crashes the entire web-app, that bug would take precedence over all other bugs.

Functional Requirements: If we label bugs under the same category of severity, we will look at our functional requirements and fix bugs that occur on P0 (must have) features over P1 (should have) features.

We will retest the solution by using our existing Jest tests (unit tests) to see if our solutions have satisfied the functional requirements. After completing bug fixes, we'll also have each team member walk through the web-app and test the features that we fixed. We would like to possibly conduct further usability tests before deploying the app again.

<u>Known Bugs and Workarounds</u>

- Users are confused about the term CSO
    - We decided to rename it everywhere and provide brief explanations in an accordion
- Notifications were difficult to implement / cost money
    - Used another JS library called emailjs
- Mobile Layout cramps information
    - Simplify mobile layout (smaller headers, dropdown menu)
- Card information not saving properly in Firebase
    - Use a JSON file because card information is static